Available online at www.sciencedirect.com
SCIENCE DIRECT®
Applied Catalysis A: General 246 (2003) 171–181
APPLIED CATALYSIS A: GENERAL
www.elsevier.com/locate/apcata

Fischer–Tropsch synthesis: Group II alkali-earth metal promoted catalysts

Mingsheng Luo, Burtron H. Davis*

Center for Applied Energy Research, University of Kentucky, 2540 Research Park Drive, Lexington, KY 40511, USA

Received 5 October 2002; received in revised form 23 December 2002; accepted 24 December 2002

Abstract

The effects of four Group II alkali-earth metals (barium, beryllium, calcium and magnesium) and potassium promoters on iron Fischer–Tropsch synthesis (FTS) catalysts product selectivity, syngas conversion and productivity are compared to the unpromoted iron catalyst. Iron FTS catalysts promoted with Group II alkali-earth metals have lower overall FTS activities and lower alpha values than a potassium promoted iron catalyst, but higher values than an unpromoted catalyst. All Group II metal promoters yielded similar carbon utilization as the unpromoted iron catalyst, but higher than a potassium promoted catalyst regardless of the space time. Carbon dioxide rates indicate that a potassium promoted iron catalyst possesses better water-gas shift (WGS) activity than any Group II metal promoted catalyst when both are operated at high CO conversions. Ca and Mg can even suppress the WGS activity below that of unpromoted iron catalysts. Among the Group II metals, a Ba promoted catalyst has the highest WGS activity while Mg and Ca promoted catalysts showed the lowest WGS activity.

A potassium promoted catalyst generated the highest yield of CO_2 and hydrocarbon, but the lowest methane and total liquid product (C_{5+}) rate. All alkali promoted catalysts yielded higher total liquid rate, but lower gas fraction than unpromoted catalysts. Alkali-earth promoted catalysts produced a slightly higher C_2–C_4 olefin ratio, but a K promoted catalyst produced a notably higher C_2 olefin fraction than Group II metal promoted catalysts.

Keywords: Fischer–Tropsch synthesis; Iron catalyst; Promoter; Alkali-earth metals; Barium; Potassium; Beryllium; Magnesium; Calcium

1. Introduction

Fischer–Tropsch synthesis (FTS) converts a mixture of CO and H_2 that can be produced from coal or natural gas to liquid fuels. One of the advantages of FTS is that the process can be designed to produce gasoline, diesel and/or chemicals. There are two types of operating modes: low temperature FTS, which is for production of wax, and high temperature FTS, which is designed for production of gasoline and alkenes.

In the FTS process, the hydrocarbon is produced from CO and H_2, and the conversion can be expressed as:

$$(2n+1)H_2 + nCO \rightarrow C_nH_{2n+2} + nH_2O$$

When an iron catalyst is used for the FTS reaction, the water-gas shift (WGS) reaction also occurs. This side reaction consumes CO and water, produced in the FTS process, to produce additional hydrogen:

$$H_2O + CO \rightarrow CO_2 + H_2$$

Among the potential FTS catalysts, such as ruthenium, nickel, iron and cobalt, only the last two are

* Corresponding author. Tel.: +1-859-257-0251; fax: +1-859-257-0302.
E-mail address: davis@caer.uky.edu (B.H. Davis).

0926-860X/03/$ – see front matter
doi:10.1016/S0926-860X(03)00024-3

commercially used in FTS processes [1–4]. Silica is believed to be an effective structure promoter for iron catalysts [5–7]; silica can also have chemical impact on catalyst properties [8–10]. Potassium has long been used as a promoter for iron catalysts to increase the alkene yield and to decrease the amount of CH_4 in the products [1,4,11]. It is believed that potassium can facilitates the dissociation of CO and promote the growth of hydrocarbon chains [1,7,12–15]. It is also believed that potassium can increase both FTS and water-gas shift catalytic activity [11,16–22]. In spite of the widespread use of potassium as a promoter, little work has been reported to allow a direct comparison of promoters at medium or high pressure FTS conditions. We have recently compared potassium to other Group I alkali-earth metal promoters under medium pressure conditions appropriate for slurry reactor operations [23]. The promotion effect of the Group I alkali metals was also studied earlier by other researchers [1,2,24,25]. The present work extends the work with Group I metals to include promoters in Group II of the Periodic Table.

Promoted iron FTS catalysts with Fe:alkali-earth metal:Si = 100:1.44:4.6 atomic ratio of Group II elements (beryllium, magnesium, calcium and barium) were used in this study. The influence of alkali-earth on product selectivity, productivity, water-gas shift and FTS activity are compared with results obtained from FTS reactions over unpromoted and potassium-promoted iron catalysts.

2. Experimental

2.1. Preparation of alkali-earth metal promoted catalysts

Four iron catalysts promoted with Be, Ba, Ca and Mg and one with potassium were prepared and tested in this study. An atomic ratio of 1.44:100 of promoter:iron was used. A precipitated iron catalyst base was prepared using tetraethyl orthosilicate and iron nitrate. A ferric nitrate solution was prepared by dissolving $Fe(NO_3)_3{\cdot}9H_2O$ in distilled and deionized water, and then the amount of tetraethyl orthosilicate needed to make Si:Fe of 4.6:100 atomic ratio was added. The mixture was stirred vigorously until the tetraethyl orthosilicate hydrolyzed. The tetraethyl orthosilicate and iron nitrate mixture was then added to a continuous stirred tank reactor (CSTR) precipitation vessel together with a stream of concentrated ammonium hydroxide. By maintaining the slurry pH at 9 and an average residence time of 6 min, a base catalyst material with an iron to silicon atomic ratio of 100:4.6 was obtained. A vacuum drum filter was used to collect the precipitate which was then washed twice with deionized water. The final filter cake was dried in an oven with flowing air at 110 °C for 24 h. The catalyst was crushed to approximately 60 μm and calcined for 4 h in a 350 °C oven under an air flow. The unpromoted iron catalyst with a Fe:Si = 100:4.6 was thus prepared and used in FTS.

In this study, the unpromoted iron catalyst was impregnated with an aqueous nitrate solution of barium, beryllium, calcium, magnesium or potassium to obtain the promoted catalysts (work should not be undertaken with beryllium before becoming familiar with proper safeguards due to its toxicity). Alkali-earth metal or potassium solution was added to obtain a promoter to iron atomic ratio of 1.44:100. Following the impregnation, the catalyst was dried at 110 °C overnight with good mixing. Thus, catalysts promoted with four Group II alkali-earth metals and one with potassium were prepared for FTS studies.

2.2. In situ catalysts activation

Prior to the synthesis operations, the iron catalyst needs to be activated with either H_2, CO or synthesis gas. Activation procedures can have a significant effect on the selectivity and activity of iron catalysts [1–3,16,27–29]. Anderson found that at 0.01 MPa and 325 °C, pure CO was a better reduction gas than syngas [26]. It was reported that catalysts activated with CO yielded more long-chain hydrocarbons than syngas and H_2 activated catalysts [28–30]. In addition, activation conditions may also influence the performance of the iron catalyst. In this study, the potassium promoted iron catalyst was pretreated with 24 SL/h g-Fe CO at 270 °C and 1.3 MPa for 24 h. The reduction of Fe_2O_3 with CO occurs in two steps:

$$3Fe_2O_3 + CO \rightarrow 2Fe_3O_4 + CO_2$$

$$5Fe_3O_4 + 32CO \rightarrow 3Fe_5C_2 + 26CO_2$$

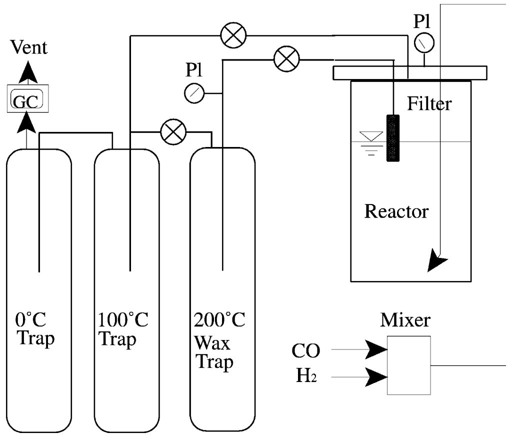

Fig. 1. Schematic of the CSTR Fischer–Tropsch reactor system.

In addition, CO_2 may be formed by the Boudouard reaction:

$$2CO \rightarrow C + CO_2$$

Previous work in our laboratory showed that, following a 24 h activation, approximately 50–100% more carbon was present in the catalyst mass than was needed to form Fe_5C_2 [26]. Gradual oxidation of unpromoted Fe_5C_2 to Fe_3O_4 was observed when the iron catalyst was used in FTS. The results showed that at the end of the run an unpromoted iron catalysts had been completely oxidized to Fe_3O_4.

2.3. Reactor system

A 1 L CSTR was used in this study (Fig. 1). A sintered metal filter was installed to remove wax samples from the catalyst slurry in the reactor to be collected in a hot trap (200 °C). The components in the vapor space of the reactor continuously passed to a warm trap (100 °C) and cold trap (0 °C) that were used to collect samples of oil, light wax and water. Tail gas from the cold trap was reduced to atmospheric pressure and analyzed with an HP quick GC.

After the catalyst was activated with CO, syngas was introduced at a rate of 10 SL/h g-catal. (SL = l at 25 °C). CO and H_2 were introduced, using mass controllers, to a gas mixing vessel to provide a simulated synthesis gas. This flow rate was maintained until the activity became constant and was used as the standard reaction condition. The gas space velocity was varied over the range of 3–60 SL/h g-Fe to collect kinetic data. At intervals, the standard conditions were re-established to obtain a conversion to establish the impact of catalyst deactivation. Other reaction conditions were maintained at 270 °C, 1.3 MPa and a stirrer speed of 750 rpm.

Table 1
Analyzers for FTS products

Analyzer	Sample	GC detector
HP Quad Series Micro GC	Gas	TCD
HP 5890A Series II	Water	TCD
Agilent 6890	Oil + light wax	FID
HP 5890 Series II Plus	Heavy wax	FID

2.4. Product sampling and analysis

Gas, water, oil, light and heavy reactor wax samples were collected daily and analyzed. Table 1 gives the summary of the instruments for gas and liquid product analysis. A heavy wax sample was taken from the 200 °C hot trap connected to the filter. Vapor phase above the slurry phase passed the warm (100 °C) and the cold (0 °C) traps outside the reactor. The light wax and water mixture was collected from the warm trap and an oil plus water sample from the cold trap. Tail gas from the cold trap was analyzed with an online HP Quad Series Micro GC and molar compositions of C_1–C_7 olefins and paraffins were obtained. Hydrogen and carbon monoxide conversions and CO_2 production were calculated based on the gas product online GC analysis results and the gas flow measured at the reactor outlet. Hydrogen, carbon monoxide and syngas conversion were obtained using the following formula:

$$\text{Conversion} = \frac{N_{\text{in}} - N_{\text{out}}}{N_{\text{in}}} \times 100\%$$

The oil and light wax samples were mixed before being analyzed with an HP 5790A GC. The heavy wax was analyzed with an HP5890 Series II Plus GC while the water sample was analyzed with an HP5890 GC.

3. Results and discussion

3.1. Effect on Fischer–Tropsch synthesis activity

For a synthesis gas of H_2:CO = 0.7:1, Fig. 2 shows the results of CO conversion over the catalysts

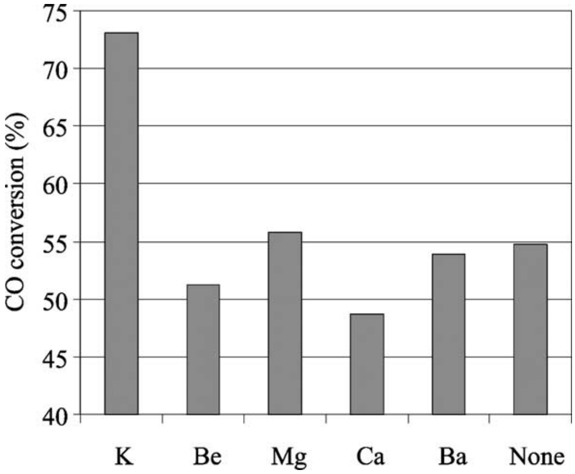

Fig. 2. Effect of promoter on iron catalyst activity for CO conversion (270 °C, 1.3 MPa and 10 SL/h g-Fe).

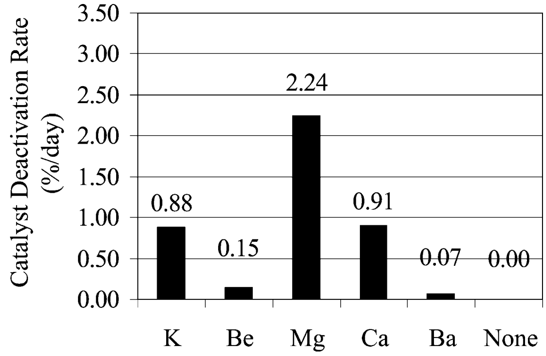

Fig. 3. Deactivation rate for iron catalysts (270 °C, 1.3 MPa and 10 SL/h g-Fe).

promoted with Group II alkali-earth metals (Ba, Be, Ca and Mg) and potassium. The potassium promoted catalyst yielded the highest overall CO conversion of 73% at the base reaction conditions of 270 °C, 1.3 MPa and 10 SL/h g-Fe. The calcium and beryllium promoted FTS catalysts yielded the lowest initial conversion of 47 and 52%, respectively. Ba and Mg promoted catalysts produced CO conversions similar to the unpromoted catalyst. This result suggests that none of the alkali-earth metals enhanced the FTS activity to values greater than the unpromoted catalyst and that Be and Ca even had a negative effect on FTS activity.

Deactivation rates (% per day) for these five catalysts were obtained from the CO conversion results over 400–500 h of FTS reaction. As described in the experimental section, FTS base reaction conditions were established and then the space velocity was changed from 3 to 50 SL/h g-Fe before the reaction conditions were switched back to the original standard one. Overall deactivation rate was calculated based on the total reaction time; therefore, the overall deactivation rate may be higher than during a run without reaction condition changes. As indicated in Fig. 3, beryllium and barium promoted catalysts yielded a low deactivation rate (0.07 and 0.15% per day, respectively), while potassium and calcium promoted catalysts deactivation rates were 0.88 and 0.91, respectively. Magnesium promoted catalysts yielded the highest deactivation rate (2.24% per day) among all five catalysts tested. For potassium, the deactivation rate is larger than when the same catalyst is used at one flow rate (3.1 SL/h g-catal.) and a high (90%) initial CO conversion where the rate is 0.15% per day or less. Presumably altering the reaction conditions causes a higher deactivation rate but this remains to be verified. The unpromoted catalyst showed a negligible CO conversion decline rate in 400 h of reaction period, suggesting that promotion with alkali and alkali-earth metals increased the initial activity resulting in a higher deactivation rate. This contrasts to results obtained with unpromoted iron catalysts operated to produce higher CO conversion levels.

Fig. 4 shows the results of CO conversions at various space velocities (10, 30 and 50 SL/h g-Fe) following establishment of stable activity at the base condition. At a medium to high space velocity, all promoted catalysts, except Ca, produced a CO conversion slightly lower than the unpromoted catalyst. At low space velocity, however, the alkali-earth promoted catalysts yielded CO conversions similar to the unpromoted catalyst. The potassium promoted catalyst yielded the highest overall CO conversion at a high space velocity. Calcium promoted catalysts produced a CO conversion lower than the unpromoted catalyst regardless the space velocity.

3.2. *Effect on Fischer–Tropsch synthesis selectivity*

Figs. 5 and 6 show the rates of hydrocarbon, methane, CO_2 and water formation for FTS reactions catalyzed with Group II alkali-earth metal and

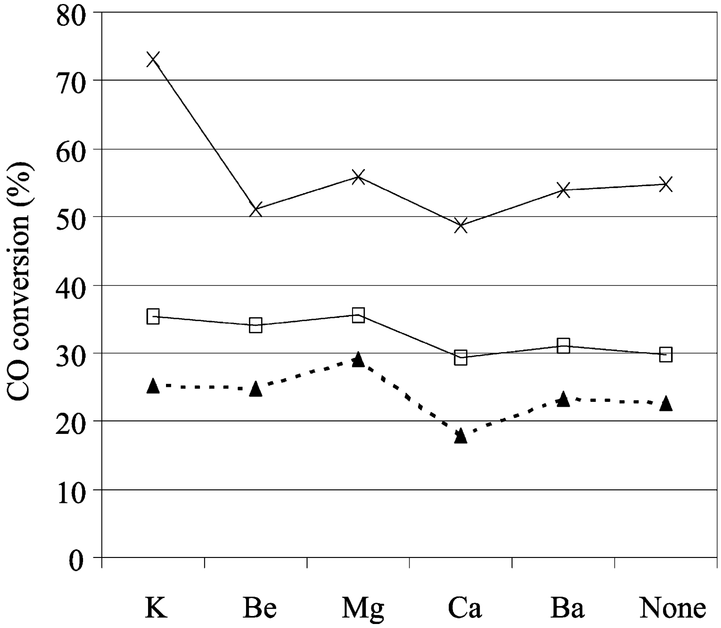

Fig. 4. Effect of promoter on CO conversions at these space velocities (270 °C, 1.3 MPa) (×, S.V. = 10; □, SV = 30; ▲, S.V. = 50 SL/h g-Fe).

potassium promoted catalysts under the same CO conversion (20%). All Group II alkali-earth metal promoted catalysts produced a higher hydrocarbon rate at this CO conversion than the unpromoted catalyst. A 3.2–3.7 g/h g-iron hydrocarbon rate was obtained for Group II alkali-earth metal promoted catalysts while the potassium promoted catalyst had a hydrocarbon rate of 2.75 g/h g-iron, which is even lower

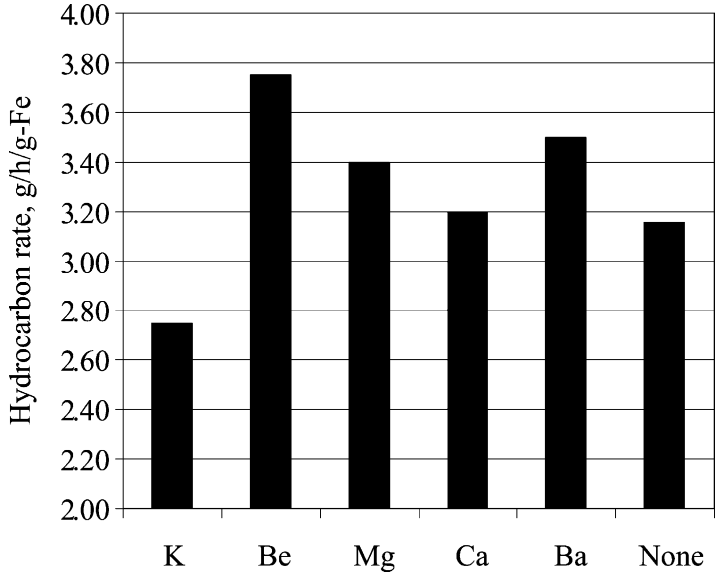

Fig. 5. Hydrocarbon rate at 20% CO conversion.

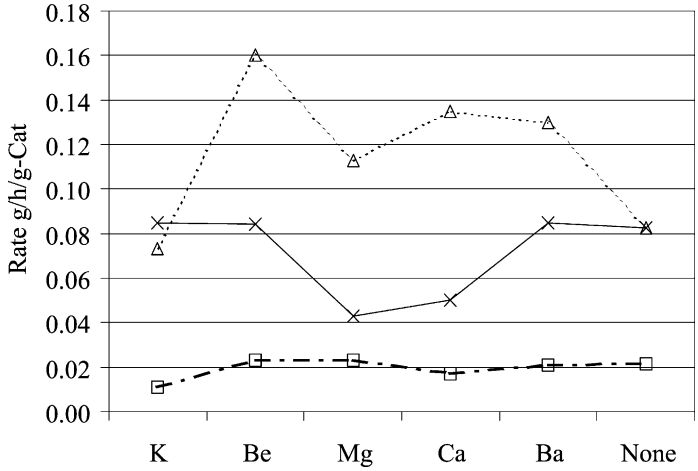

Fig. 6. Production of CO_2, H_2O and CH_4 at 20% CO conversion (×, CO_2; △, H_2O; □, CH_4).

than the unpromoted catalyst. This rate depends on selectivity so that, at a CO conversion level, the poorest WGC catalyst will have the highest hydrocarbon rate. Fig. 6 shows that the rate for methane production for the potassium promoted catalyst was lower than all Group II alkali-earth metal promoted and the unpromoted catalysts. While catalysts promoted with K, Ba and Be produced a CO_2 rate similar to the unpromoted catalyst, Mg and Ca yielded a lower CO_2 rate than unpromoted catalyst. The potassium promoted catalyst yielded water and CO_2 rates lower than the unpromoted iron catalyst. All alkali-earth promoted iron catalysts generated a higher water rate than the unpromoted catalyst. The data in Fig. 6 also show that calcium and magnesium promoted catalysts produced lower CO_2 rates than the unpromoted catalyst, indicating that both Mg and Ca can suppress the WGS activity.

Olefin ratio was calculated from the following equation:

$$\text{olefin ratio} = \frac{\text{olefin}}{\text{olefin} + \text{paraffin}}$$

Fig. 7 shows the olefin ratio from FTS over potassium and Group II alkali-earth metal promoted iron catalysts compared at the same 20% CO conversion. The data show that all Group II alkali-earth metal and potassium promoted catalysts yielded similar C_3 and C_4 olefin ratios that fall in the 0.85–0.90 range, slightly higher than unpromoted catalyst. The ethylene fraction, however, shows a significant difference between potassium and Group II alkali-earth metal promoted catalysts. An ethene fraction of 0.76 was obtained for the potassium promoted catalyst while an ethene fraction of 0.30–0.35 was obtained for the Group II alkali-earth metal promoted catalysts, showing that potassium suppressed the hydrogenation activity better than all alkali-earth metals. All promoted catalysts produced a higher C_2 olefin ratio than unpromoted catalyst.

The distribution of hydrocarbon fractions (Fig. 8) show that the potassium promoted catalyst produced more gas (C_1–C_4) but lower gasoline (C_5–C_{11}) than the Group II alkali-earth metal promoted catalysts

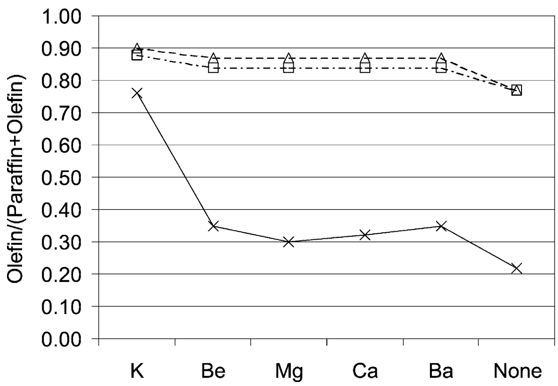

Fig. 7. Effect of alkali promoters on olefin ratios at 20% CO conversion (×, C_2; △, C_3; □, C_4).

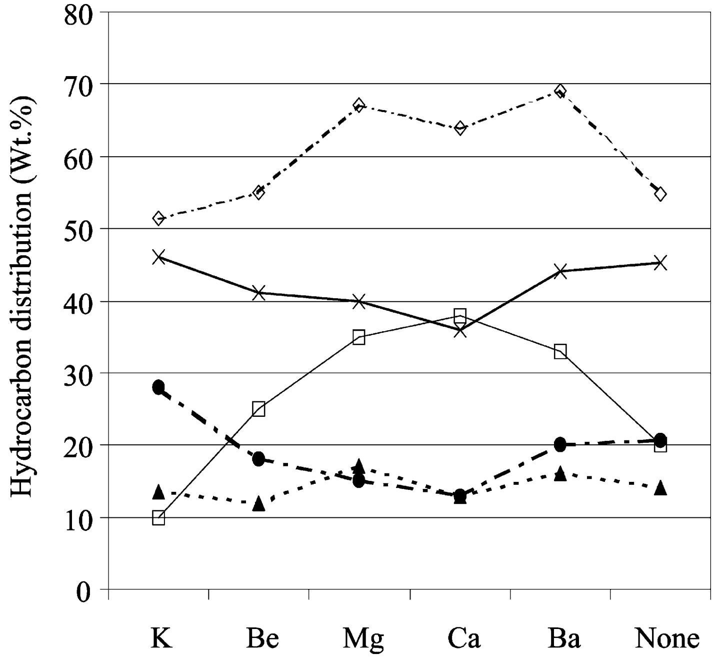

Fig. 8. Effect of promoters on hydrocarbon distribution at 20% CO conversion (×, C_1–C_4; □, C_5–C_{11}; ▲, C_{12}–C_{18}; ●, C_{19+}; ◇, C_{5+}).

and the unpromoted catalyst. All promoted catalysts produced a similar C_{12}–C_{18} fractions to each other. Among the Group II alkali-earth metal promoters, Ca produced the largest gasoline fraction (44.9%) and the smallest gas (C_1–C_4) and C_{19+} fraction while the Ba promoted catalyst generated the highest C_{19+} and C_1–C_4 fractions. In summary, all alkali-earth promoted catalysts yielded a higher liquid (C_{5+}) but lower gas fractions than unpromoted catalyst, while the potassium promoted catalyst produced a slightly higher gas and lower total liquid product than unpromoted catalyst. These data provide trends but an accurate product distribution can be obtained only after long times of operation at one set of conditions.

Fig. 9 shows the CO_2 selectivity obtained from FTS reactions catalyzed by Group II alkali-earth metal promoted catalysts. In this study, space time was varied in the range of 0.02–0.33 SL/h g-Fe), (i.e. space velocities in the range of 3–50 SL/h g-Fe). The potassium promoted catalyst produced more CO_2 than any alkali-earth metal promoted catalyst in the whole range of space time although the CO_2 selectivity trends converge at a space time of about 0.1. The unpromoted iron catalyst produced a CO_2 selectivity that is similar to alkali-earth promoted catalysts at a low space time, but this increased rapidly to a level higher than all Group II alkali-earth and potassium promoted catalysts at a space time above about 0.1 SL/h g-Fe. Magnesium and calcium promoted catalysts produced a lower CO_2 yield than beryllium and barium promoted catalysts. At a low space velocity, the barium promoted catalyst generated the most CO_2 while all the other Group II alkali-earth metal promoted catalysts produced a similar CO_2 selectivity (40–42%). Therefore, calcium and magnesium promoted catalysts gave better carbon utilization than beryllium and barium promoted catalysts. The potassium promoted catalysts produced a lower carbon utilization than any of the Group II alkali-earth metal promoted catalysts.

As another measure of carbon utilization, methane selectivity is shown in Fig. 10. Although the potassium promoted catalyst produced the most CO_2, it generated the least methane at high space velocity among the six catalysts tested in this study. The unpromoted catalyst produced the highest CH_4 selectivity at a low space time. Except at low space velocity, all Group II alkali-earth metal promoted catalysts generated similar amounts of methane.

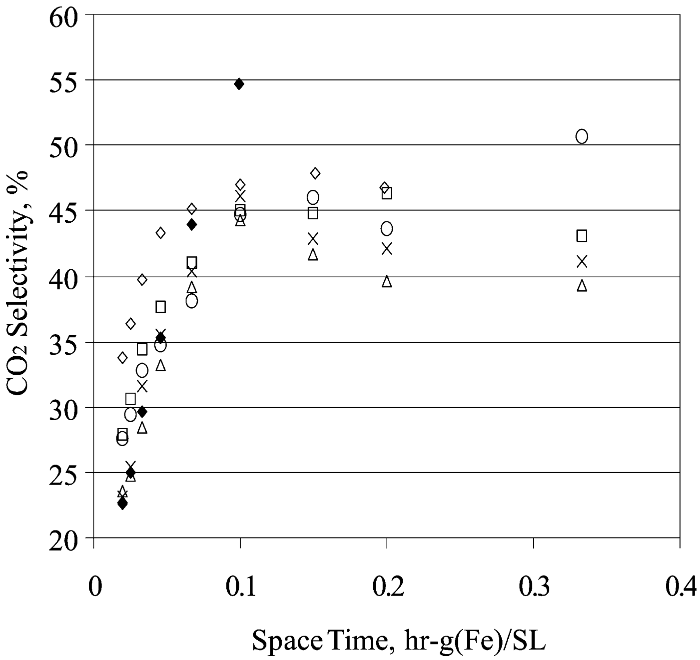

Fig. 9. Influence of alkali promoter on CO_2 selectivity (◇, K; □, Be; △, Mg; ×, Ca; ○, Ba; ◆, none).

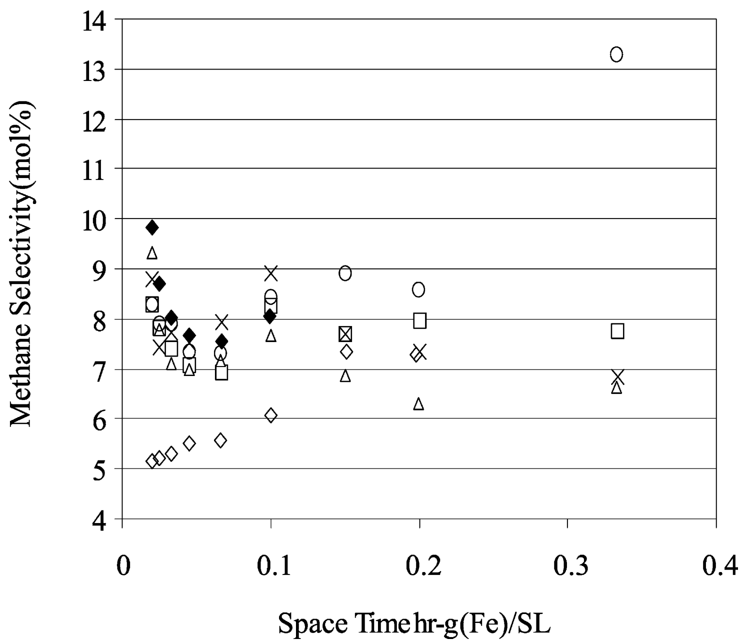

Fig. 10. Dependency of methane selectivity on promoter and space velocity (◇, K; □, Be; △, Mg; ×, Ca; ○, Ba; ◆, none).

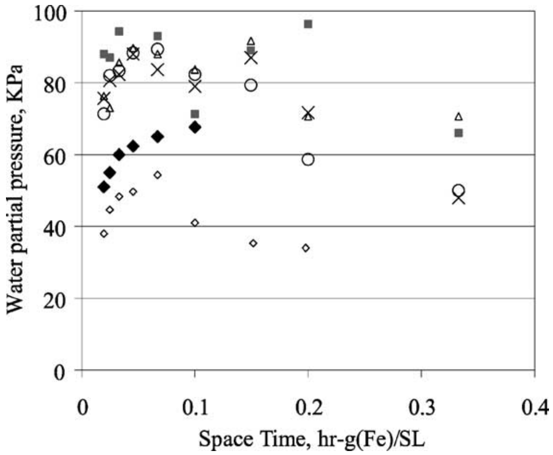

Fig. 11. Effect of promoter and space time on the water partial pressure in the reactor velocity (◇, K; ■, Be; △, Mg; ×, Ca; ○, Ba; ◆, none).

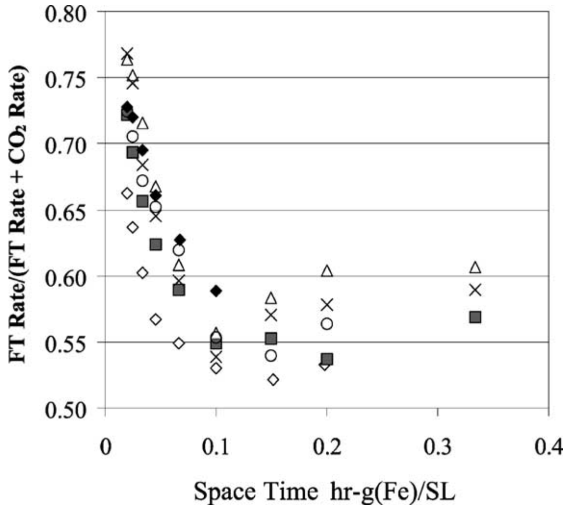

Fig. 12. Effect or promoter and space time on carbon utilization (◇, K; ■, Be; △, Mg; ×, Ca; ○, Ba; ◆, none).

Over the range of CO conversion (space time range), the potassium promoted catalyst provided a lower water partial pressure than the other promoted catalysts (Fig. 11). All Group II alkali-earth promoters generated a higher water partial pressure than unpromoted iron catalyst. Even so, all of the promoted catalysts produced the maximum water partial pressure at about the same space time (0.05; 40–50% CO conversion).

The carbon usage ratio (fraction of CO converted to hydrocarbons) was lower for the promoted catalysts than the unpromoted catalyst. The potassium

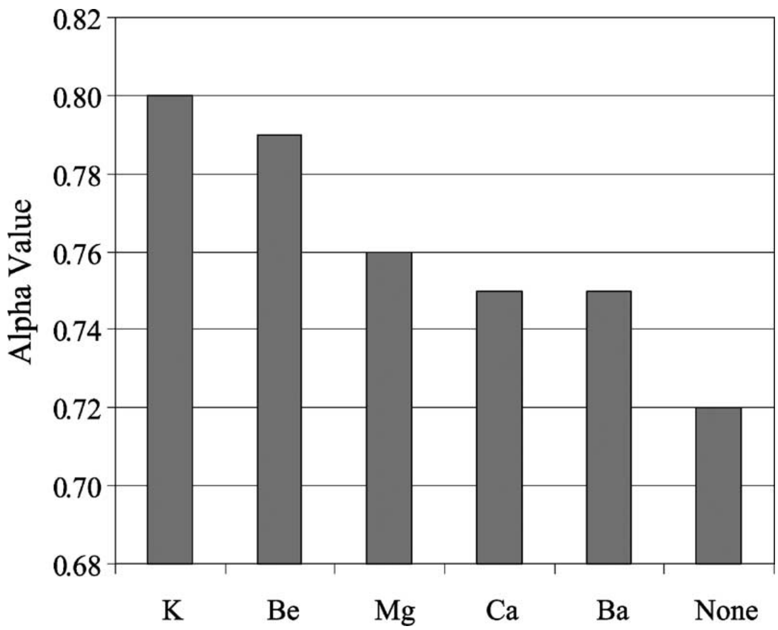

Fig. 13. Effect of alkali promoter on alpha values.

promoted catalyst produced the lowest carbon usage among all promoted catalysts. The utilization ratio for the four alkali-earth promoted catalysts are similar (Fig. 12).

The alpha values obtained for the promoted catalyst are summarized in Fig. 13. The potassium promoted catalyst had the largest alpha value, 0.80. There was a gradual decline in alpha for the alkali-earth promoted catalysts as the promoter was located lower in the Periodic Table, attaining 0.75 as the lowest value. All promoters enhanced the alpha value in comparison to the unpromoted catalyst.

4. Conclusion

Iron FTS catalysts promoted with Group II alkali-earth metals produced lower overall FTS activity and lower alpha values than a potassium promoted iron catalyst. All alkali-earth promoters produced values similar or even slightly lower CO conversion than the potassium promoted catalysts at base FTS condition of 270 °C, 1.3 MPa and 10 SL/h g-Fe. The catalytic activity of the potassium promoted catalyst declined more rapidly when the conditions were changed than when operating at the same temperature, pressure and 3.1 NL/h g-Fe (90% CO conversion initially). All promoters generated a higher FTS alpha value than the unpromoted catalyst. The results also showed a gradual decline in alpha for the alkali-earth promoted catalysts as the promoter was located lower in the Periodic Table. The lowest alpha values for the promoted catalysts were obtained for the Ba and Ca promoted catalysts. Among the Group II alkali-earth metals, Mg and Ba promoters caused slightly higher FTS activity than Be and Ca promoted catalysts. Ba and Be promoted catalysts showed better stability than the potassium and calcium promoted catalysts which were more stable than the magnesium promoted catalyst. All Group II alkali-earth metal promoted catalysts yielded a higher carbon utilization than the potassium promoted catalyst but had a carbon utilization rate similar to the unpromoted catalyst regardless of the space time.

Carbon dioxide production shows that the potassium promoted catalyst possesses better WGS activity than any Group II alkali-earth promoted catalyst. Both Mg and Ca showed a negative effect on WGS activity as a lower CO_2 rate was obtained from the Mg and Ca promoted catalysts than from the unpromoted catalyst.

Acknowledgements

This work was supported by US DOE contract number DE-FC26-98FT40308 and the Commonwealth of Kentucky.

References

[1] M. E. Dry, in : J. Anderson, M. Boudart (Eds.), Catalysis Science and Technology, vol. 1, Springer, Berlin, 1981.
[2] H.H. Storch, N. Golumbia, R.B. Anderson, The Fischer–Tropsch and Related Synthesis, Wiley, New York, 1951.
[3] R.B. Anderson, The Fischer–Tropsch Synthesis, Academic Press, New York, 1984.
[4] R.J. O'Brien, L. Xu, R.L. Spicer, B.H. Davis, in: Proceedings of the Symposium on Syngas Conversion to High Value Chemicals at the 211th ACS Annual Meeting, New Orleans, 24–29 March, 1996, pp. 252–253.
[5] V.U.S. Rao, G.J. Steigel, G.J. Cinquegrane, R.D. Srivastava, Fuel Proc. Technol. 30 (1992) 83.
[6] M.E. Dry, in: Bruce E. Leach (Ed.), Applied Industrial Catalysis, vol.2, Academic Press, NY, 1983, p. 167.
[7] C.D. Frohning, in: J. Falbe (Ed.), New Synthesis with Carbon Monoxide, Springer, New York, 1980, p. 353.
[8] C. Li, Effect of potassium and copper promoters on reduction behavior of precipitated iron catalysts, Ph.D. Thesis, Texas A&M University, 1988.
[9] J.L. Rankins, C.H. Bartholomew, J. Catal. 100 (1986) 533.
[10] J.-A. Dalmon, R. Dutartre, G.A.C.R. Martin, Acad. Sci. Ser. C 287 (1978) 557.
[11] R.A. Dictor, A.T. Bell, J. Catal. 97 (1986) 121–136.
[12] A.G. Sault, J. Catal. 140 (1993) 136–149.
[13] D.J. Dwyer, J.H. Hardenburg, Appl. Surf. Sci. 19 (1984) 14.
[14] H.P. Bonzel, H.J. Kreb, Surf. Sci. 109 (1981) l527.
[15] D.A. Wesner, F.P. Coenen, H.P. Bonzel, Langmuir 1 (208) (1985) 478.
[16] D.B. Bukur, D. Mukesh, S.A. Patel, Ind. Eng. Chem. Res. 29 (1990) 194.
[17] M.E. Dry, T. Shingles, L.J. Boshoff, G.J. Oosthuizen, J. Catal. 15 (1969) 190.
[18] J. Benziger, R.J. Madix, Surf. Sci. 94 (1980) 119.
[19] V. Arakawa, A.T. Bell, I&EC Proc. Des. Dev. 22 (1993) 97.
[20] M. Luo, B.H. Davis, in: Studies in Surface Science and Catalysis, vol. 139, Elsevier, Amsterdam, 2001, p. 133.
[21] R.J. O'Brien, L. Xu, R.L. Spicer, B.H. Davis, Energy Fuels 10 (1996) 921.
[22] M. Luo, R.J. O'Brien, S. Bao, B.H. Davis, Appl. Catal. A: Gen. 239 (2002) 111.

[23] N.H. Wilfried, Y. Zhang, R.J. O'Brien, M. Luo, B.H. Davis, Appl. Catal. A: Gen. 236 (2002) 77.
[24] M.E. Dry, G.J. Oosthuizen, J. Catal. 11 (1968) 18.
[25] M. Röper, in: W. Keim (Ed.), Catalysis in C_1 Chemistry, Reidel, Dordrecht, 1983, pp. 41–88.
[26] R.B. Anderson, in: P.H. Emmett (Ed.), Catalysis, vol. 4, Reinhold, New York, 1958, p. 29.
[27] A.G. Sault, J. Catal. 140 (1993) 121–135.
[28] R.J. O'Brien, Y. Zhang, H.H. Hamdeh, B.H. Davis, ACS Div. Pet. Chem. 44 (1999) 100–102, 21–25 March 1999, Anaheim, CA (preprints).
[29] M. Luo, B.H. Davis, Fuel Processing Technology, submitted, 2002.
[30] D.B. Bukur, M. Koranne, X. Lang, K.R.P.M. Rao, G.P. Huffman, Appl. Catal. A: Gen. 126 (1995) 85.